FlowGuard

A Multi-Agent Digital Twin System for Real-World Crowd Management

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Presented by

MANISHA M 2116240701304

MATHU METHAA K 2116240701306



Problem Statement & Abstract

Problem Statement

Large gatherings at stadiums, religious sites and public venues face recurring crowd-safety risks during entry, exit and emergencies. Today's options are manual stewarding with no simulated support, or single-venue commercial twins that need heavy sensor infrastructure.

Abstract

FlowGuard builds a multi-agent digital twin for any location picked on a map. It pulls real gates, footpaths and roads from OpenStreetMap, estimates crowd capacity from polygon area and satellite imagery, and lets classical AI agents decide how to route and respond.

BDI Agents

Crowd, Route and Safety agents perceive the twin and form goals.

Utility Coordinator

Scores candidate plans on evacuation time, density and safety.

Dijkstra & A*

Real shortest and safest paths on the OpenStreetMap graph.

CSP Solver

Assigns crowd groups to exits under capacity and distance limits.

No ML or DL: every decision traces back to a rule, a score or a search algorithm

Literature Survey

Author & Year	Methodology	Drawback	Improvement in FlowGuard
Li et al. (2021)	Survey of sensor, feature and decision-level data fusion for crowd monitoring.	Descriptive only; no routing or decision-making.	A working decision pipeline (agents + Coordinator), not just a taxonomy.
Noor (2023)	Hidden Markov Model classifies CCTV footage into normal or abnormal flow (Hajj/Umrah).	Needs dense CCTV; detects problems but decides no response.	No cameras needed; map data plus agent-driven rerouting.
Li, S. (2022)	IoT-based stadium management system architecture.	No AI decision layer; one fixed venue.	Works for any venue and adds the missing AI decision layer.
Mu (2023)	1,500+ WiFi access points used as passive sensors on a campus.	Privacy-sensitive device tracking; observation only.	Uses public map data and adds real intervention.
Zhong et al. (2024)	Four-layer digital twin for airport crowd control.	Airport-specific; not shown to generalize.	Same twin-to-decision layering, but location-independent.

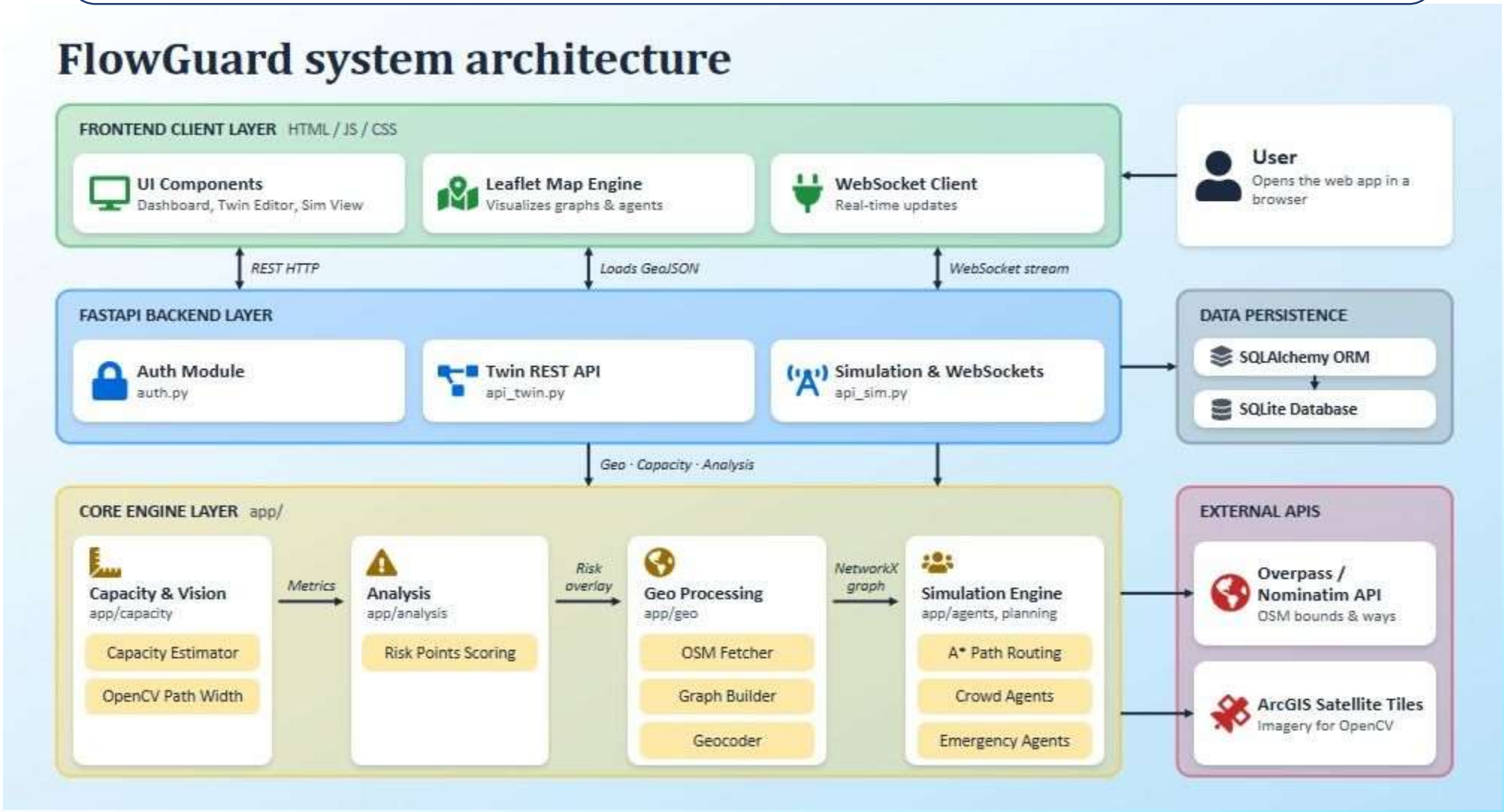
Common thread (1/2): the studies either monitor crowds without deciding what to do, or lean on hardware that is tied to one venue.

Literature Survey

Author & Year	Methodology	Drawback	Improvement in FlowGuard
Yasufuku & Takahashi (2024)	Agent-based simulation platform predicting congestion 10-20 minutes ahead.	Their own analysis: linear prediction misses crowd dynamics.	No numeric prediction; the Coordinator scores explicit plans.
Meng et al. (2022)	A* with path-density and obstacle cost for multi-storey ship evacuation.	Ship-specific; one algorithm, no multi-agent coordination.	Density-aware routing inside a full multi-agent system with CSP.
Golubić (2023)	Review of Dijkstra, A* and linear programming for evacuation planning.	Theory only; no implementation or real data.	Implemented and tested on real stadium graphs with live simulation.
Dinh Thi Hong Huyen et al. (2023)	Dijkstra with route-capacity limits and blocked-route re-planning.	Capacity is assumed; no estimation or agent coordination.	Estimates capacity (OSM + image analysis); adds CSP and agents.
Chen & Zhang (2025)	4K cameras, infrared, UWB and MobileNetV3 edge computing for a smart gym.	Heavy hardware; cannot scale to a new venue.	Zero hardware; any venue is available from OpenStreetMap.

FlowGuard closes all three gaps: real OpenStreetMap data (no hardware), capacity estimation, and BDI multi-agent coordination, for any location.

SYSTEM ARCHITECTURE



Implementation

User
Selects a location on the map

→

Frontend
Leaflet map, HTML / CSS / JS

→

Backend
FastAPI, JWT, WebSocket

→

Twin Builder
OSM graph + OpenCV capacity

→

Agents
BDI + Coordinator + CSP

→

Live Map
Crowd, routes, risk layers

Stack & Setup

- Python: FastAPI, SQLite, NetworkX
- OpenStreetMap Overpass + Nominatim geocoding
- OpenCV HSV segmentation for capacity
- Dijkstra, A*, priority queue, CSP backtracking
- Scenarios: blocked gate, zone incident, inflow surge, combined crisis

Results

3
real Tamil Nadu venues tested

506 – 1,085
graph nodes per twin

~35,600
average estimated capacity

12
risk points flagged

Agent-driven rerouting cut average evacuation time and peak zone density versus an unmanaged baseline.

Modules

MODULE 1

Authentication & Twin Management

JWT login with bcrypt hashing; users save, view and delete their own twins.

MODULE 2

Location-Based Twin Generation

Turns an address or map click into a NetworkX graph of real gates, paths and roads.

MODULE 3

Capacity Estimation

Polygon area plus OpenCV satellite analysis gives a comfortable / dense / danger range.

MODULE 4

Multi-Agent Crowd Management

BDI agents perceive the twin; a utility Coordinator picks the best action plan.

MODULE 5

Risk Analysis & Emergency Response

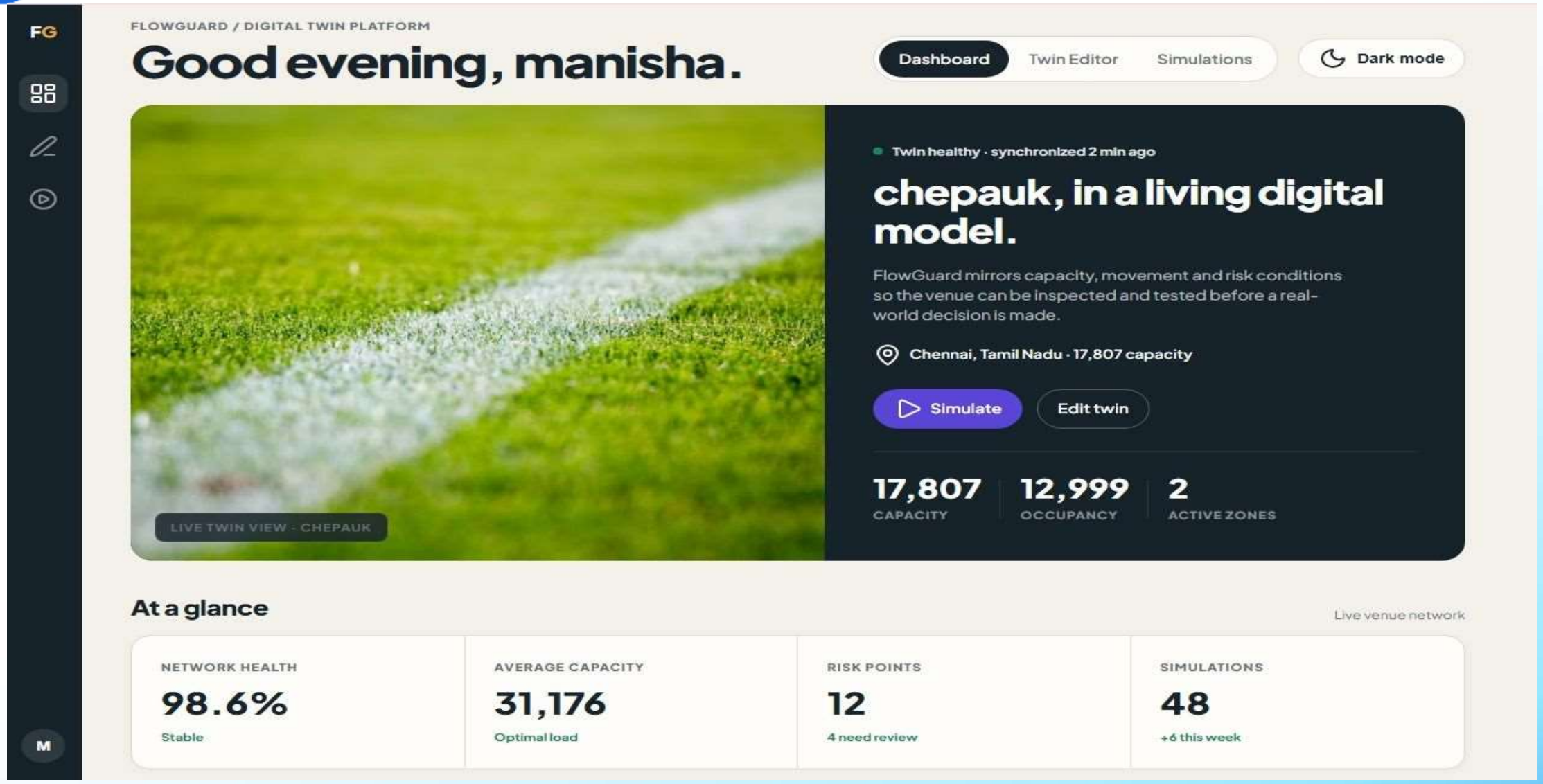
Flags bottlenecks, narrow paths and stairs; routes responders and ranks hospitals.

MODULE 6

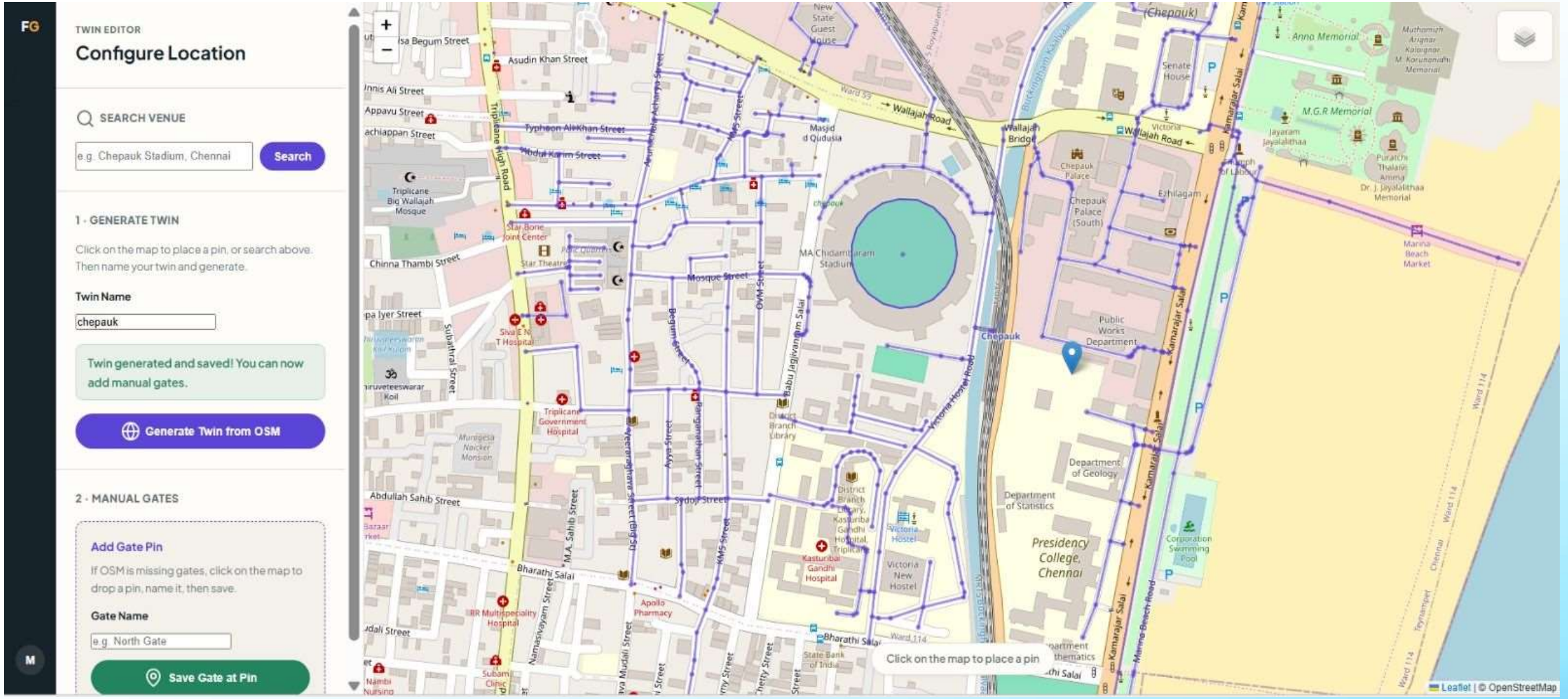
Simulation & Visualization

Tick-based live simulation on a Leaflet map with a readable agent-reasoning trace.

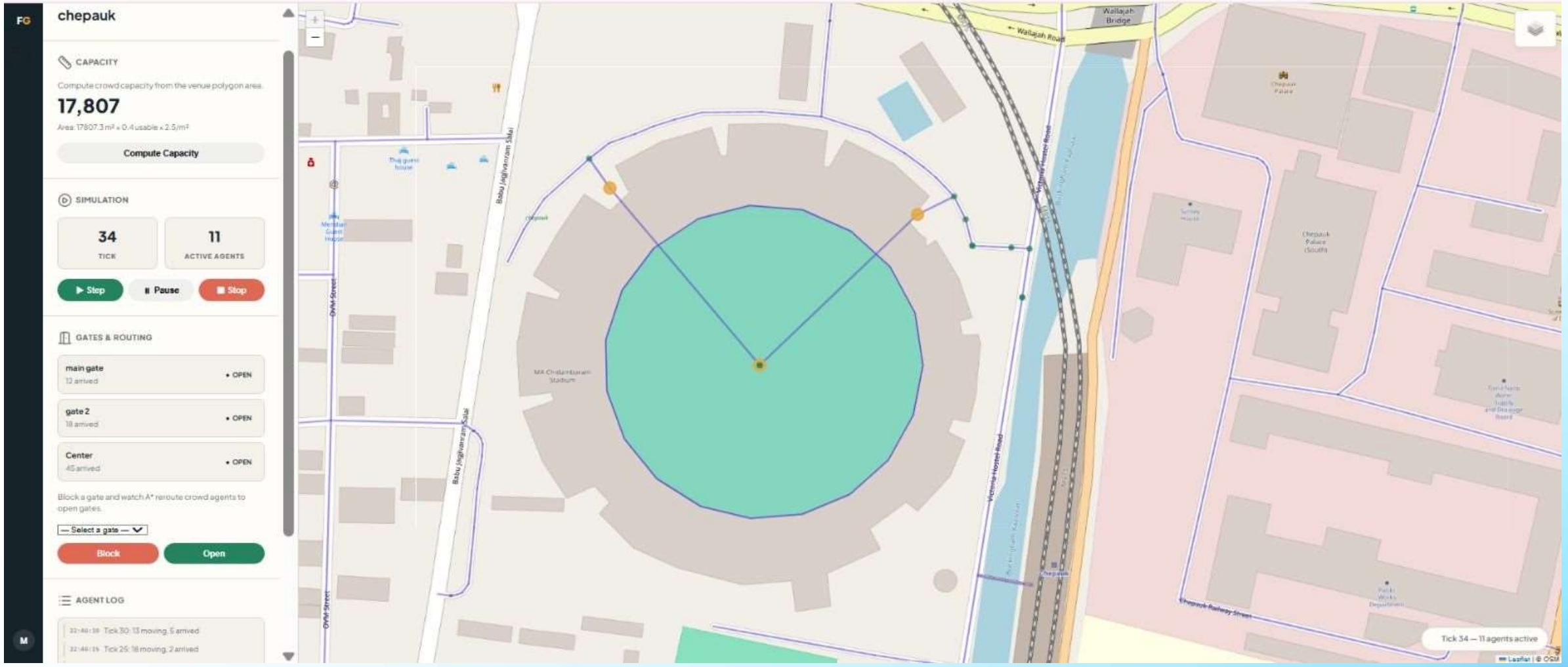
RESULTS


FG
FLOWGUARD / DIGITAL TWIN PLATFORM
Good evening, manisha.
Dashboard
Twin Editor
Simulations
Dark mode
LIVE TWIN VIEW - CHEPAUK
Twin healthy · synchronized 2 min ago
chepauk, in a living digital model.
FlowGuard mirrors capacity, movement and risk conditions so the venue can be inspected and tested before a real-world decision is made.
Chennai, Tamil Nadu · 17,807 capacity
Simulate
Edit twin
17,807
CAPACITY
12,999
OCCUPANCY
2
ACTIVE ZONES
At a glance
Live venue network
NETWORK HEALTH
98.6%
Stable
AVERAGE CAPACITY
31,176
Optimal load
RISK POINTS
12
4 need review
SIMULATIONS
48
+6 this week
M

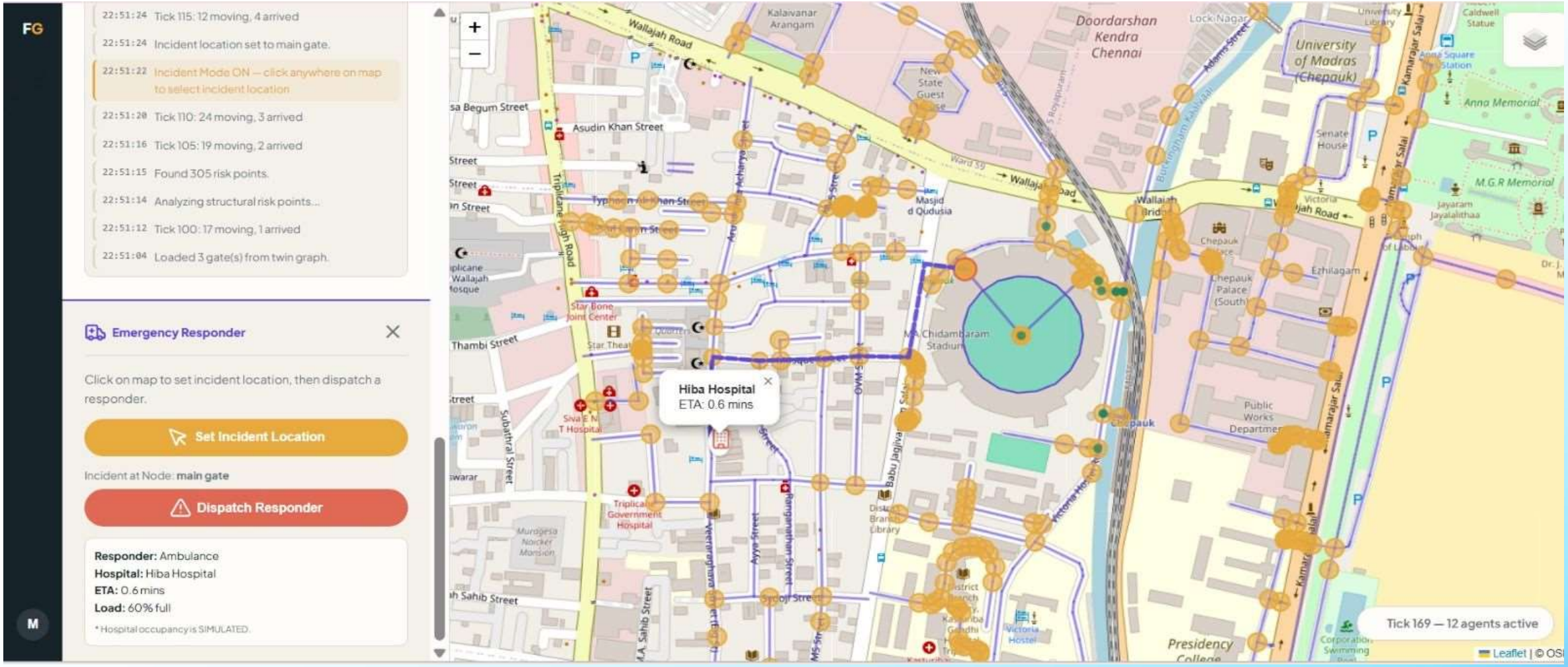
RESULTS


FG
TWIN EDITOR
Configure Location
SEARCH VENUE
e.g. Chepauk Stadium, Chennai
Search
1 - GENERATE TWIN
Click on the map to place a pin, or search above. Then name your twin and generate.
Twin Name
chepauk
Twin generated and saved! You can now add manual gates.
Generate Twin from OSM
2 - MANUAL GATES
Add Gate Pin
If OSM is missing gates, click on the map to drop a pin, name it, then save.
Gate Name
e.g. North Gate
Save Gate at Pin
M
Click on the map to place a pin
Leaflet | © OpenStreetMap

RESULTS


FG
chepauk
CAPACITY
Compute crowd capacity from the venue polygon area.
17,807
Area: 17807.3 m² × 0.4 usable × 2.5/m²
Compute Capacity
SIMULATION
34
TICK
11
ACTIVE AGENTS
▶ Step
⏸ Pause
■ Stop
GATES & ROUTING
main gate
12 arrived
• OPEN
gate 2
18 arrived
• OPEN
Center
45 arrived
• OPEN
Block a gate and watch A* reroute crowd agents to open gates.
— Select a gate —
Block
Open
AGENT LOG
Tick 30: 13 moving, 5 arrived
Tick 25: 18 moving, 2 arrived
M
Wallajah Bridge
Wallajah Road
Victoria Hostel Road
Tick 34 — 11 agents active

RESULTS


FG
22:51:24 Tick 115: 12 moving, 4 arrived
22:51:24 Incident location set to main gate.
22:51:22 Incident Mode ON — click anywhere on map to select incident location
22:51:20 Tick 110: 24 moving, 3 arrived
22:51:16 Tick 105: 19 moving, 2 arrived
22:51:15 Found 305 risk points.
22:51:14 Analyzing structural risk points...
22:51:12 Tick 100: 17 moving, 1 arrived
22:51:04 Loaded 3 gate(s) from twin graph.
Emergency Responder
Click on map to set incident location, then dispatch a responder.
Set Incident Location
Incident at Node: main gate
Dispatch Responder
Responder: Ambulance
Hospital: Hiba Hospital
ETA: 0.6 mins
Load: 60% full
* Hospital occupancy is SIMULATED.
M
Hiba Hospital
ETA: 0.6 mins
Tick 169 — 12 agents active
Leaflet | © OS

Conclusion & Future Enhancements

Conclusion

- One pipeline adapts to any real venue; it worked on three Tamil Nadu stadiums without venue-specific tuning.
- Combines OpenStreetMap data, image-based capacity, BDI agents, a utility Coordinator, Dijkstra / A* and CSP.
- Fully explainable: no ML or DL, every decision is a visible rule, score or search.
- Agent rerouting measurably lowered evacuation time and peak density against an unmanaged baseline.

Future Enhancements

- Transparent, non-ML demand forecasting from holiday, festival and weather data.
- Visitor-facing SMS / app notifications.
- Live road-traffic integration for responder routing.
- Validation against publicly reported real-world congestion incidents.
- Negotiation-based coordination between agents with competing goals (e.g. Route vs Parking Agent).

References

1. Li, X. et al. (2021) 'Data Fusion for Intelligent Crowd Monitoring and Management Systems: A Survey', IEEE Access, Vol. 9.

2. Noor, T.H. (2023) 'Behavior Analysis-Based IoT Services for Crowd Management', The Computer Journal, 66(9), pp. 2208-2220.

3. Li, S. (2022) 'Development of Intelligent Sports Stadium System Based on Internet of Things Technology', Proc. CTMCD 2022.

4. Mu, M. (2023) 'Network as a Sensor for Smart Crowd Analysis and Service Improvement', IEEE Network, 37(2), pp. 144-152.

5. Zhong, J. et al. (2024) 'Intelligent Airport Crowd Management Technology Based on Digital Twin', J. System Simulation, 36(1), pp. 27-38.

6. Yasufuku, K. and Takahashi, A. (2024) 'Development of a Real-Time Crowd Flow Prediction and Visualization Platform for Crowd Management', J. Disaster Research, 19(2), pp. 248-255.

7. Meng, D., Hu, Z. and Zhang, H. (2022) 'Simulation of Multi-layer Ship Evacuation System Based on Improved A* Algorithm', J. System Simulation, 34(6), pp. 1375-1382.

8. Golubić, I. (2023) 'Mathematical Evacuation Planning: Graph Theory and Optimization Approaches', DKU Journal, Veleučilište Velika Gorica.

9. Dinh Thi Hong Huyen et al. (2023) 'Solving Optimization Problems in Emergency Evacuation', J. Science and Technology, Univ. of Danang, 6.2.

10. Chen, J. and Zhang, D. (2025) 'Design of IoT-Based Crowd Flow Monitoring System for Smart Sports Venues', Advances in Transdisciplinary Engineering, 70.

11. Dijkstra, E.W. (1959) 'A note on two problems in connexion with graphs', Numerische Mathematik, 1, pp. 269-271.

12. Rao, A.S. and Georgeff, M.P. (1995) 'BDI Agents: From Theory to Practice', Proc. ICMAS-95.

13. Russell, S. and Norvig, P. (2021) Artificial Intelligence: A Modern Approach, 4th edn, Pearson.

14. OpenStreetMap contributors (2024) OpenStreetMap, openstreetmap.org (accessed 2026).

15. Hagberg, A., Swart, P. and Chult, D.S. (2008) 'Exploring network structure, dynamics, and function using NetworkX', SciPy 2008, pp. 11-15.